

Chapter 16

Acid-Base Equilibria

1. Which one of the following is a Brønsted-Lowry acid? E

A) $(\text{CH}_3)_3\text{NH}^+$ ✓

B) CH_3COOH ✓

C) HF ✓

D) HNO_2 ✓

E) all of the above

Brønsted–Lowry (Acid-base proton theory):

- 1) Acid is a substance that donates a proton to another substance;
- 2) Base is a substance that accepts a proton from another substance;

2. Which one of the following statements regarding K_w is false?

B

- A) pK_w is 14.00 at 25 °C. ✓
- B) The value of K_w is always 1.0×10^{-14} . ✗
- C) K_w changes with temperature. ✓
- D) K_w is known as the ion product of water. ✓

Ion-product constant for water:

$$K_w = [H_3O^+][OH^-] = [H^+][OH^-] = 1.0 \times 10^{-14} \quad (T=25^\circ\text{C})$$

3. Which one of the following is the weakest acid? D

A) HF ($K_a = 6.8 \times 10^{-4}$)

B) HClO ($K_a = 3.0 \times 10^{-8}$)

C) HNO₂ ($K_a = 4.5 \times 10^{-4}$)

D) HCN ($K_a = 4.9 \times 10^{-10}$)

E) CH₃COOH ($K_a = 1.8 \times 10^{-5}$)

2) Acid-dissociation constant: $K_a = [H^+][A^-]/[HA]$;

3) The greater the value of K_a , the stronger is the acid;

4. Which of the following acids will be the strongest? A

5. The K_a of hypochlorous acid (HClO) is 3.0×10^{-8} at 25°C . What is the percent ionization (%) of hypochlorous acid in a 0.015 M aqueous solution of HClO at 25°C ? **D**



A) 4.5×10^{-8}

B) 14

C) 2.1×10^{-5}

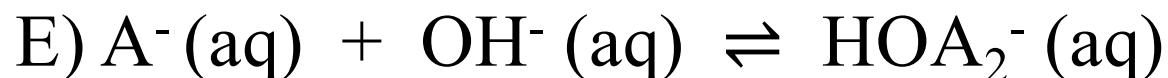
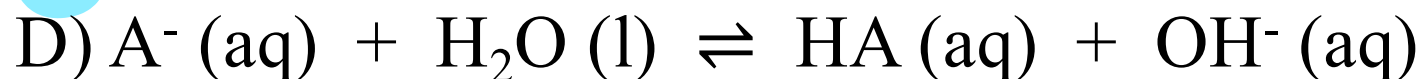
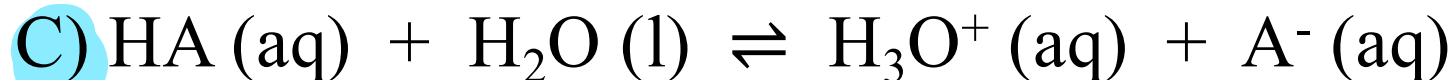
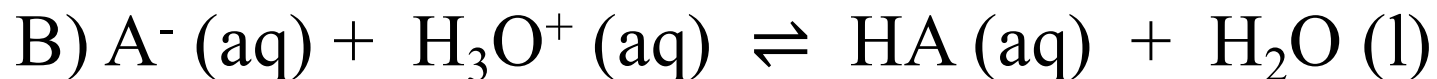
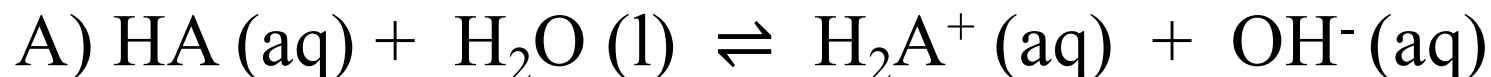
D) 0.14

E) 1.4×10^{-3}

$$\frac{X^2}{0.015} = 3 \times 10^{-8}$$

$$X = 2.12 \times 10^{-5}$$

6. A^- is a weak base. Which equilibrium corresponds to the equilibrium constant K_a for HA? C



react with water to form $OH^-(aq)$,

Weak acids: $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

1) Weak acids only partially ionized to ions in aqueous solution;

2) **Acid-dissociation constant: $K_a = [H^+][A^-]/[HA]$;** 酸常数

7. Using the data in the table, which of the conjugate acids below is the strongest acid? C

A) HClO

B) HCO_3^-

C) H_2S K_a ပိုကြီး (K_b ပိုသေး)

D) NH_3CH_3^+

E) H_2S and HClO

Base	K_b
ClO^-	3.3×10^{-7}
CO_3^{2-}	1.8×10^{-4}
HS^-	1.8×10^{-7}
NH_2CH_3	4.4×10^{-4}

8. Which of the following aqueous solutions has the highest $[\text{OH}^-]$?

D

pH 12.0

A) a solution with a pH of 3.0

B) a 1×10^{-4} M solution of HNO_3 pH = 4

C) a solution with a pOH of 12.0 pH = 2

D) pure water pH = 7

E) a 1×10^{-3} M solution of NH_4Cl
pH 5.0

10. Of the compounds below, a 0.1 M aqueous solution of A will have the highest pH. K_a လေးခုပေါ်မှာပဲ

A) KCN, K_a of HCN = 4.0×10^{-10}

B) NH_4NO_3 , K_b of NH_3 = 1.8×10^{-5} $K_a = 5.6 \times 10^{-10}$

C) NaOAc, K_a of HOAc = 1.8×10^{-5}

D) NaClO, K_a of HClO = 3.2×10^{-8}

E) NaHS, K_b of HS^- = 1.8×10^{-7}

$$K_a = 5.6 \times 10^{-8}$$